

Green Hydrogen Pilot – Feasibility Study

5 MW PEM electrolyser at Port of Eastholm with Arcturus Steel AG · Final report

Sofie Lindqvist · 28 August 2026 · H2-ENG-RPT-0001

Executive summary

Weyland Energy and Arcturus Steel AG have assessed the feasibility of a **5 MW PEM electrolyser** at the Port of Eastholm, producing green hydrogen for reheating furnace trials at Arcturus's Eastholm rolling mill, 1.4 km from the port. The study ran from March to August 2026 and was completed on **28 August 2026**.

The pilot is technically feasible and has no safety showstoppers. In the base case it produces **481 t of hydrogen per year** at a levelised cost of hydrogen (LCOH) of **EUR 9.72/kg**. With an EU Innovation Fund grant of EUR 10.5 m the LCOH falls to **EUR 7.66/kg**, below the EUR 8/kg threshold set at the go/no-go gate in January 2026. The study recommends proceeding to the funding application and to FEED.

Key figure	Value
Electrolyser	5 MW PEM (two 2.5 MW stacks)
Hydrogen output	481 t/a (≈ 1,319 kg/day average)
Full-load hours	5,200 h/a
Specific consumption (system)	54.0 kWh/kg
Capex	EUR 21.5 m
LCOH without grant	EUR 9.72/kg
LCOH with Innovation Fund grant	EUR 7.66/kg
CO2 avoided at Arcturus	≈ 3.2 kt/a (replacing natural gas in the furnace)
Target commercial operation	Q4 2028

Table 1 – Pilot at a glance



Concept rendering of the electrolyser plant on the Eastholm quayside

1. Background and objectives

Arcturus Steel AG is the offtaker of Weyland's 15-year PPA WE-PPA-2026-004 (150 MW from Harrow Bay, signed 12 May 2026). Arcturus wants to decarbonise its rolling mill at Eastholm, where natural gas fires the reheating furnace. Hydrogen co-firing at up to 30 % (by energy) is possible with burner modifications; the pilot is sized to supply this trial phase and later a full furnace zone.

Objectives of the study: (1) confirm technical feasibility at the port site, (2) estimate LCOH and funding need, (3) select electrolyser technology, (4) identify safety and permitting requirements, and (5) define the commercial structure with Arcturus. The study team was led by Isabel Ferreira (Hydrogen Project Engineer) under the steering of Sofie Lindqvist and an Arcturus energy manager.

2. Site

The site is a 0.9 ha gravel plot on the eastern quay of the Port of Eastholm, next to the planned Harrow Bay O&M base. It has a 33 kV supply nearby, process water from the municipal network and space for a 1.4 km pipeline along the existing port service corridor to the Arcturus rolling mill.

IMAGE PLACEHOLDER

hydrogen-port-aerial

Aerial drone photograph of a mid-sized North Sea port with quays, warehouses, a few cargo ships and an empty gravel plot next to the water, a large steelworks rolling mill visible in the background, overcast daylight.

Port of Eastholm – eastern quay plot with the Arcturus rolling mill in the background

3. Hydrogen demand

Arcturus plans to use about 330 t in the trial year and around 490 t/a from year three, with lower demand in July (mill maintenance shutdown). The base case production of 481 t/a matches full operation; during the trial year surplus hydrogen will be sold to a local bus operator via tube trailer (letter of intent expected Q4 2026).

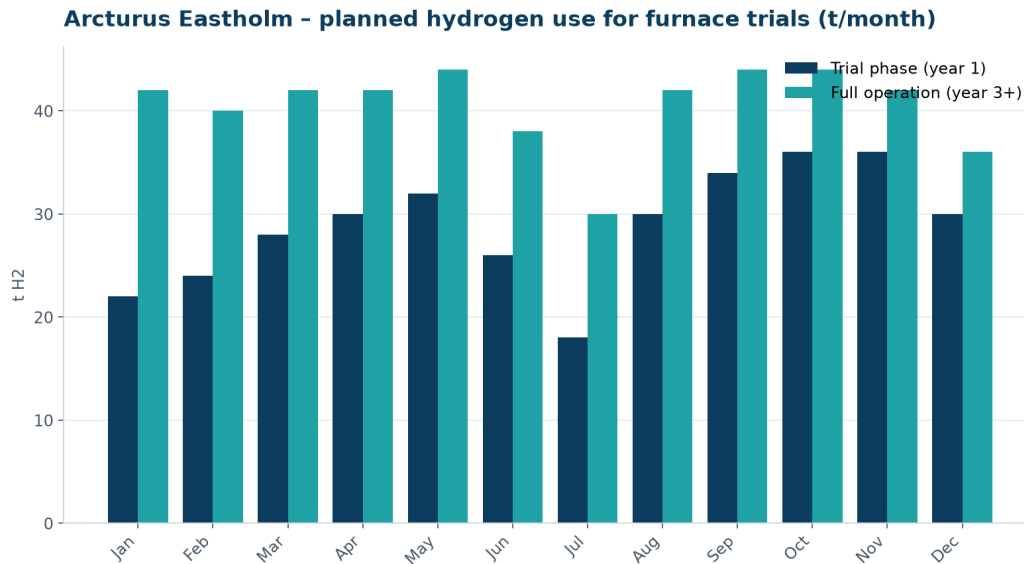


Figure 1 – Arcturus monthly hydrogen demand

4. Technology selection

Four suppliers responded to the request for information (three PEM, one alkaline). PEM was selected for its dynamic response to variable wind power, compact footprint and higher outlet pressure (30 bar). The vendor comparison is documented separately; two PEM vendors are shortlisted for FEED.

Criterion	PEM	Alkaline (pressurised)
Load range and ramp rate	5–100 %, seconds	15–100 %, minutes
Specific consumption (system, BoL)	53–56 kWh/kg	51–54 kWh/kg
Footprint for 5 MW	≈ 450 m ²	≈ 750 m ²
Stack lifetime	60,000–80,000 h	80,000–95,000 h
Critical materials	Iridium, platinum	Nickel
Capex (stack + power electronics)	Higher	Lower

5. Costs

Capex item	EUR m
Electrolyser stacks and power electronics	7.4
Balance of plant (water treatment, cooling, purification)	3.1
Compression (30 → 200 bar) and buffer storage 2.5 t	2.3
Hydrogen pipeline to Arcturus rolling mill (1.4 km)	1.3
Grid connection 33 kV and transformer	1.6
Civil works and electrolyser hall	2.2
Engineering, project management, owner's costs	1.9
Contingency (≈ 9 %)	1.7
Total capex	21.5

Table 2 – Capex estimate (AACE class 4, ±30 %)

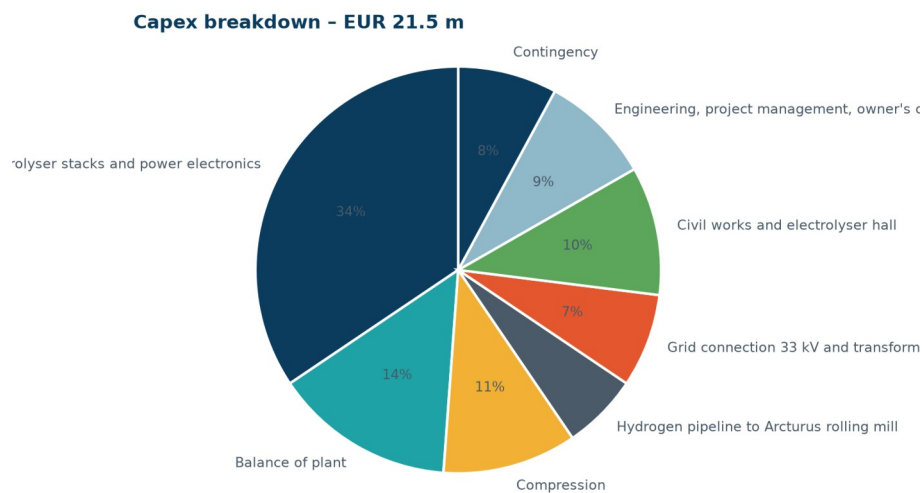


Figure 2 – Capex breakdown

Operating costs: fixed O&M of 2.5 % of capex per year, a stack replacement provision of EUR 0.22 m/a (replacement after ~70,000 h), two FTE shared with the Harrow Bay O&M base and process water. Electricity: base price EUR 62/MWh including network charges, sourced as a sleeved share of Harrow Bay output plus grid power in low-price hours (guarantees of origin for all volumes).

6. Levelised cost of hydrogen

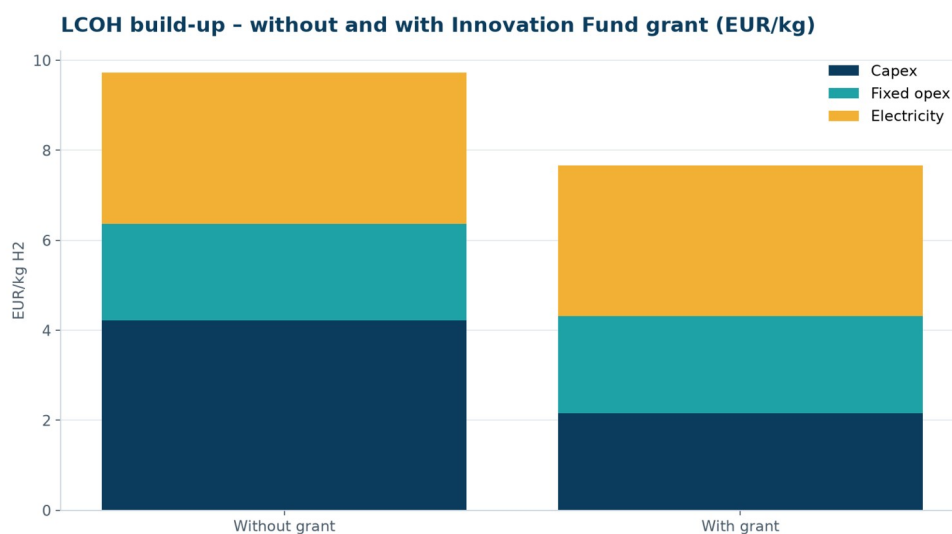


Figure 3 – LCOH build-up without and with grant

Without grant, capital cost is the largest component (4.22 EUR/kg), followed by electricity (3.35 EUR/kg) and fixed opex (2.15 EUR/kg). The grant cuts the capex share to 2.16 EUR/kg, after which electricity becomes the biggest item. The LCOH is most sensitive to full-load hours and electricity price:

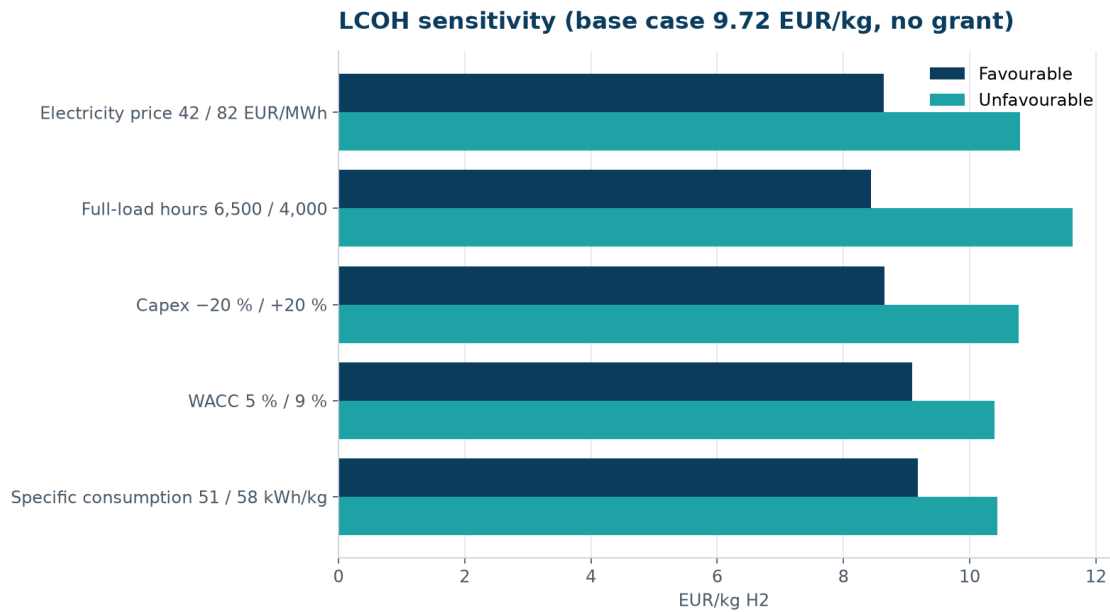


Figure 4 – LCOH sensitivities (without grant)

Sensitivity	Favourable (EUR/kg)	Unfavourable (EUR/kg)
Electricity price 42 / 82 EUR/MWh	8.64	10.80
Full-load hours 6,500 / 4,000	8.44	11.63
Capex -20 % / +20 %	8.65	10.78
WACC 5 % / 9 %	9.09	10.39
Specific consumption 51 / 58 kWh/kg	9.18	10.44

7. Safety and permitting

Hydrogen inventory is limited to 2.5 t, below the lower-tier threshold (5 t) of the major accident hazards regulations. A HAZID workshop on 17 June 2026 identified 34 hazards and no showstoppers. ATEX zoning, gas detection, an emergency shutdown system and a vent stack are part of the safety concept. Permits needed: planning consent (port development order), hazardous substances consent check, environmental permit for water discharge, and pipeline approval. Total permitting time is estimated at 12–15 months.

8. Commercial structure

- Weyland builds, owns and operates the electrolyser through a project company.
- Arcturus signs a 10-year hydrogen offtake agreement with a take-or-pay of 70 % of contracted volume.
- Price: indexed to natural gas plus a green premium, with a floor; the remaining gap to the LCOH is covered by the grant and a national carbon contract for difference (under review).
- Arcturus contributes the pipeline section on its own premises and burner modifications (not in our capex).

9. Risks

Risk	Rating	Mitigation
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Innovation Fund application not successful	High	National hydrogen support scheme as fallback; re-apply in next call
Stack degradation higher than warranted	Medium	Performance guarantee, degradation test data in tender
Arcturus demand lower than planned	Medium	Take-or-pay; second offtaker (buses) for surplus
Electricity price above EUR 75/MWh	Medium	Fixed-price sleeved supply from Harrow Bay PPA volumes
Permitting delays at port	Low	Early engagement with harbour authority; site already zoned industrial

10. Recommendations

1. Proceed to EU Innovation Fund application (deadline January 2027).
2. Launch FEED with two shortlisted PEM vendors (Q4 2026 – Q2 2027), budget EUR 1.2 m.
3. Negotiate the hydrogen offtake agreement term sheet with Arcturus by end of 2026.
4. Start permitting pre-application discussions with the harbour authority.
5. Final investment decision targeted for Q4 2027; commercial operation Q4 2028.

Prepared by Isabel Ferreira with Mei Tanaka (electrical), Oliver Grant (civil), Hannah Weber (HSE review) and Arcturus Steel's energy team. Approved by Sofie Lindqvist, Head of Engineering, 28 August 2026.